

E4_Propulsion_Exploded_Assembly: drawing created 11/30/2025 9:04:47 AM, last modified 12/2/2025 8:47:57 PM

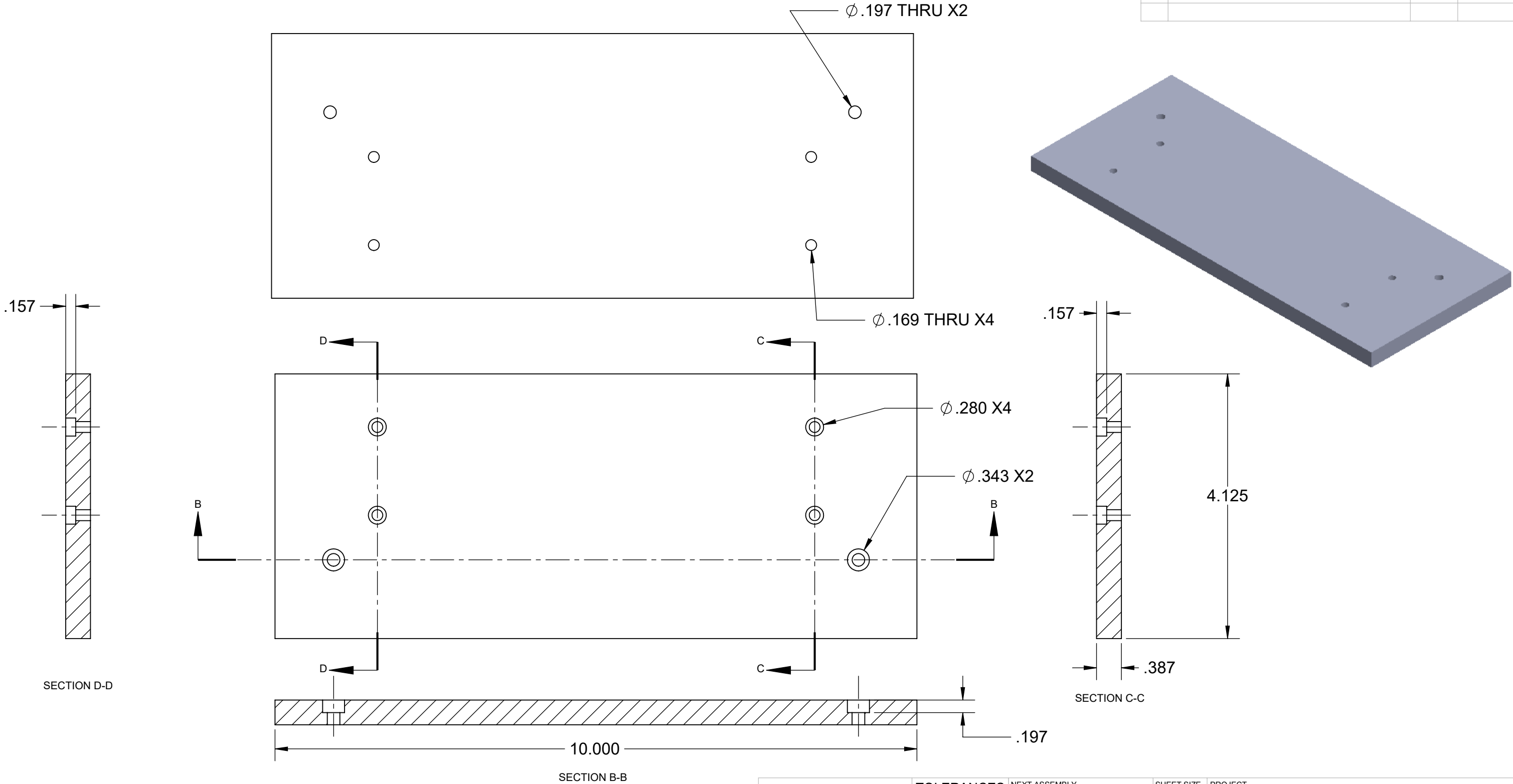
The data in this document incorporate proprietary rights of Harvey Mudd College. Any party accepting this document does so in confidence and agrees that it shall not be duplicated in whole or in part, nor disclosed to others, without the consent of Harvey Mudd College.

ITEM NO.	PART NUMBER	DESCRIPTION	Price	QTY.	Cost
1	AB1	Aluminum Base	5.00	1	5.00
2	LB1	Lego Base	12.42	1	12.42
3	T1	Tray	0.83	1	0.83
4	RS1	Right Support	1.50	1	1.50
5	76388-H	3-16in x 1in Steel Dowel Pins	0.56	2	1.12
6	SSRS7R3DDF	Ball Bearing	2.00	2	4.00
7	LS1	Left Support	1.50	1	1.50
8	PB1	Press Board	1.22	1	1.22
9	15705A45	Large Steel L Bracket	1.15	2	2.30
10	69646-F	5mm x 12mm Socket Cap Screw 0.80 Pitch	0.54	4	2.16
11	75615-N	4mm x 20mm Socket Cap Screw 0.70 Pitch	1.15	4	4.60
12	69582-B	4mm Flat Washer	0.11	4	0.44
13	63782-B	8-32 Stop Nut	0.31	4	1.24
14	61943-C	#10 3-16in Split Lock Washer	0.10	4	0.40
15	69623-C	5mm Stainless Steel Metric Hex Nuts 0.80 Pitch	0.30	4	1.20
16	62164-D	4-40 x 3-4in Zinc Slotted Round Head Machine Screw	0.11	2	0.22
17	1556A26	Mini Steel L Bracket	1.48	2	2.96
18	61941-A	#6 1-8in Split Lock Washer	0.10	2	0.20
19	62168-H	4-40 Zinc Hex Nut	0.10	2	0.20

REV	DESCRIPTION	BY	DATE
A	INITIAL RELEASE	SQ	12/02/2025

Harvey Mudd College www.HMC.edu 301 Platt Blvd Claremont, CA 91711 ©2019 ALL RIGHTS RESERVED	TOLERANCES (EXCEPT AS NOTED)	NEXT ASSEMBLY	SHEET SIZE	PROJECT	
	DIMENSIONS IN INCHES[IN]	1	B	E4_Propulsion Lab Helper	
	DECIMAL: .xx = ±.015 .xxx = ±.005	MATERIAL	DRAWING TITLE		
	TYP EDGE BREAK: .005-.015 ANGULAR: ±0.5°	FINISH	DRAWN BY		DATE
	PROJECT CODE	DRAWING NUMBER		SHEET	REV
Propulsion		EA1		1 OF 1 A	

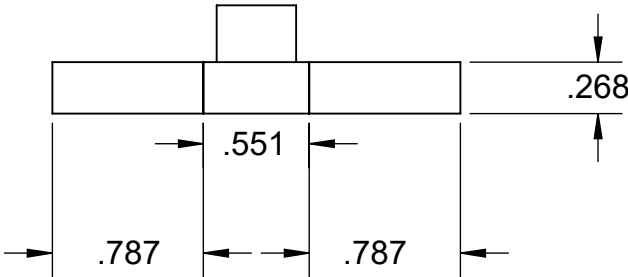
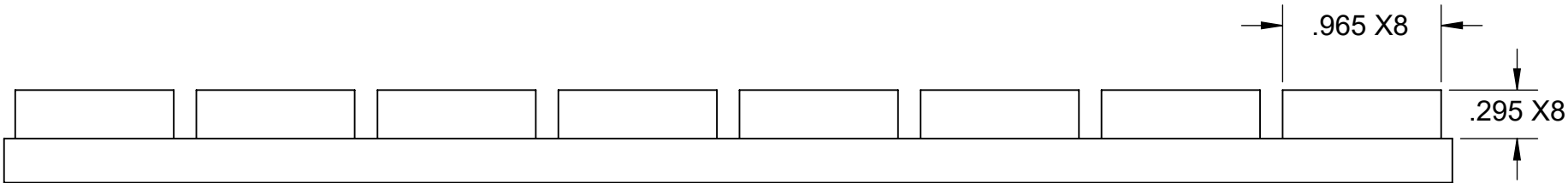
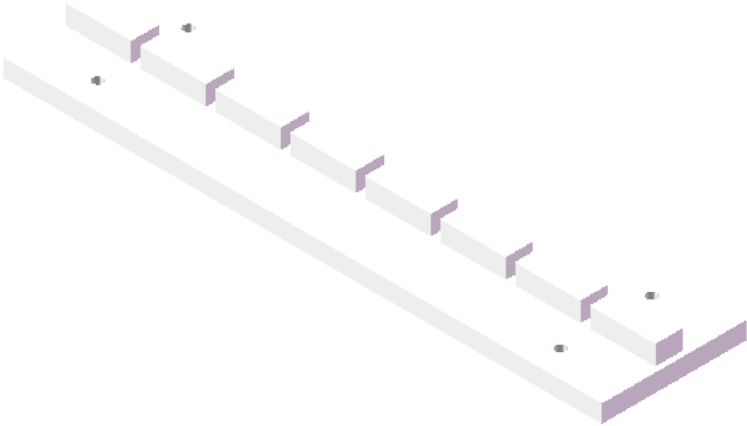
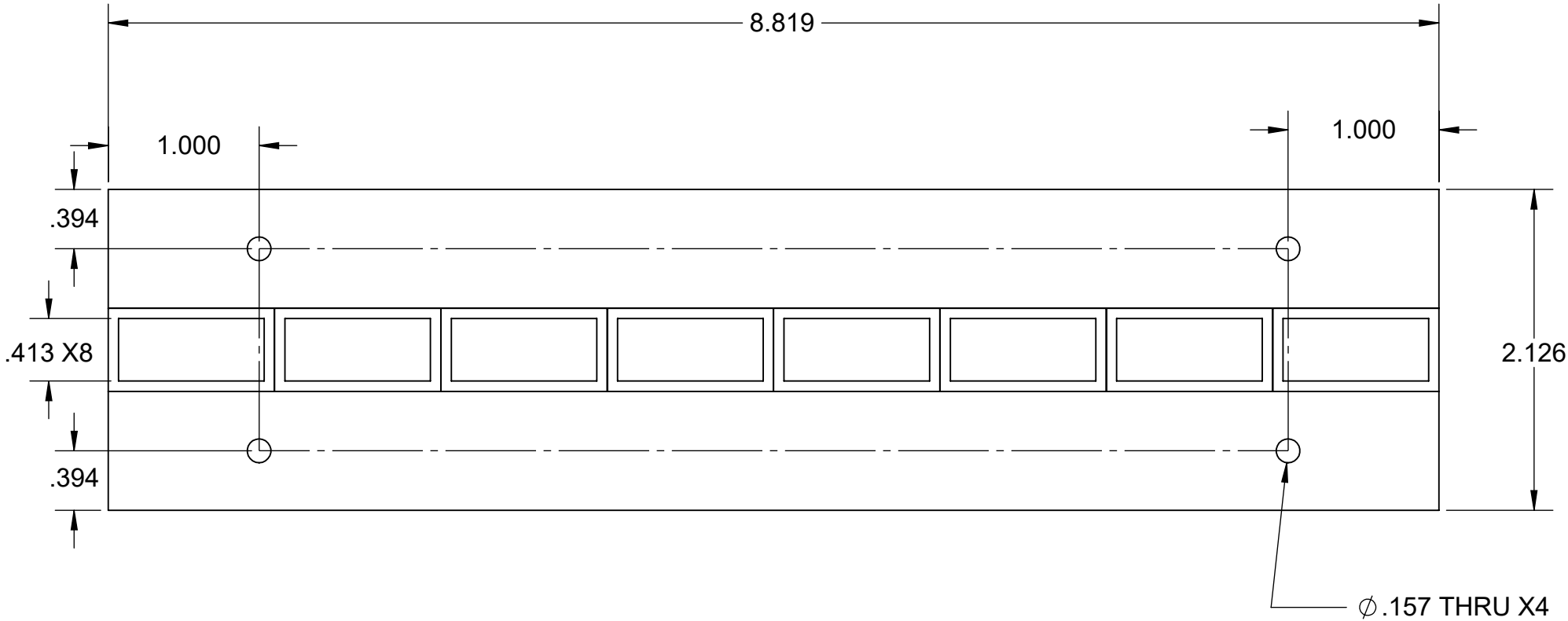
REV	DESCRIPTION	BY	DATE
A	INITIAL RELEASE	SQ	12/02/2025



Aluminum Base; drawing created 12/2/2025 8:11:18 PM, last modified 12/2/2025 8:24:00 PM

Harvey Mudd College www.HMC.edu 301 Platt Blvd Claremont, CA 91711 ©2019 ALL RIGHTS RESERVED	TOLERANCES (EXCEPT AS NOTED)	NEXT ASSEMBLY	SHEET SIZE	PROJECT		
	DIMENSIONS IN INCHES[IN]	1	B	E4_Propulsion Lab Helper		
	DECIMAL: .xxx = ±.005	MATERIAL	DRAWING TITLE			
	TYP EDGE BREAK: .005-.015	Aluminum 3003	Aluminum Base			
	ANGULAR: ±0.5°	FINISH	DRAWN BY		DATE	SHEET
		Polishing Disk	SQ		12/02/2025	1 OF 1
		PROJECT CODE	DRAWING NUMBER			REV
		Propulsion	AB1			A

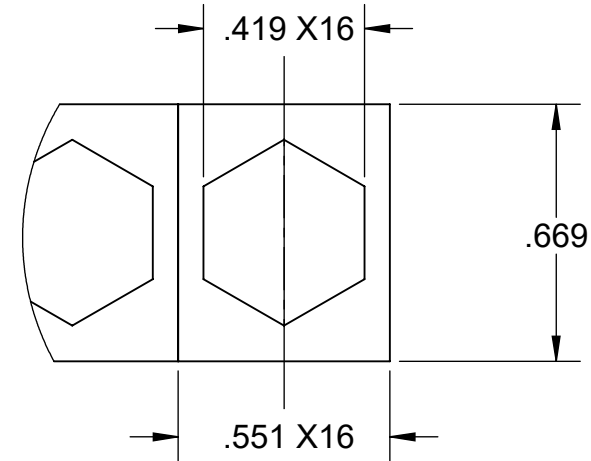
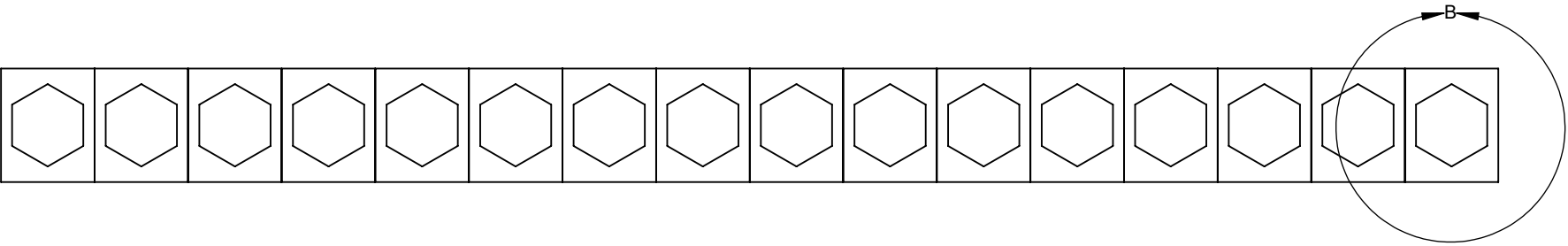
REV	DESCRIPTION	BY	DATE
A	INITIAL RELEASE	SQ	12/02/2025



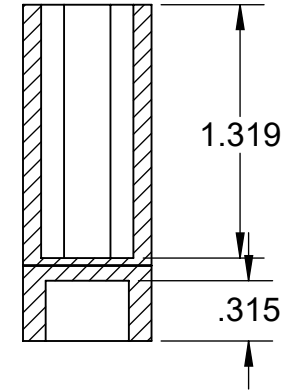
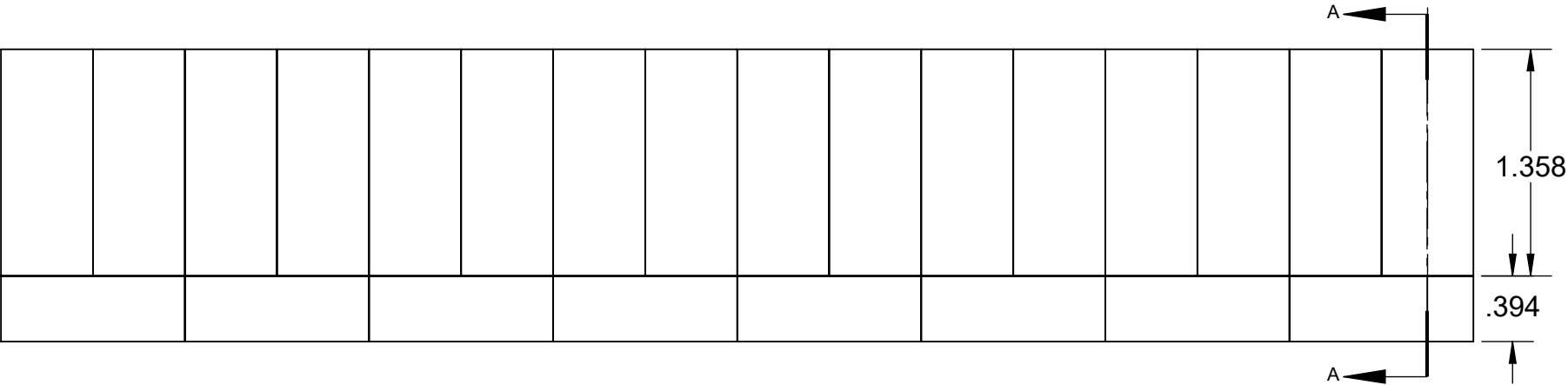
Lego Base; drawing created 11/29/2025 3:26:41 PM, last modified 12/2/2025 7:49:45 PM

Harvey Mudd College www.HMC.edu 301 Platt Blvd Claremont, CA 91711 ©2019 ALL RIGHTS RESERVED	TOLERANCES (EXCEPT AS NOTED)	NEXT ASSEMBLY	SHEET SIZE	PROJECT		
	DIMENSIONS IN INCHES[IN]	1	B	E4_Propulsion Lab Helper		
	DECIMAL: .xxx = ±.015	MATERIAL Carbon Fiber Nylon	DRAWING TITLE Lego Base			
	TYP EDGE BREAK: .005-.015	FINISH 3D Printed	DRAWN BY SQ	DATE 12/02/2025	SHEET 1 OF 1	
	ANGULAR: ±0.5°	PROJECT CODE Propulsion	DRAWING NUMBER LB 1			REV A

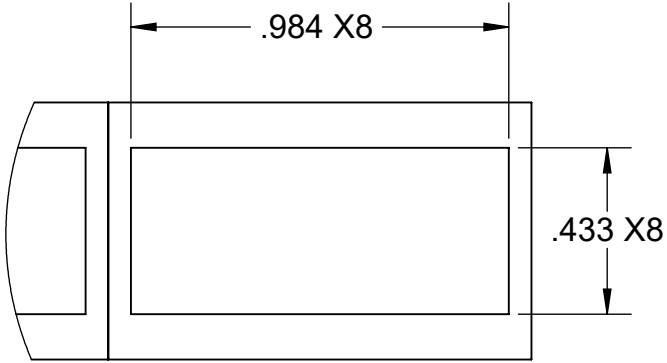
REV	DESCRIPTION	BY	DATE
A	INITIAL RELEASE	SQ	12/02/2025



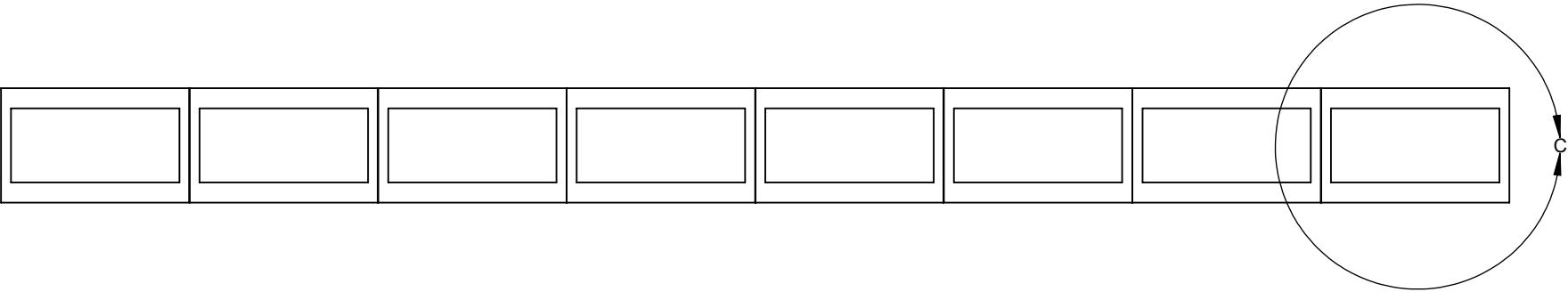
DETAIL B



SECTIONA-A

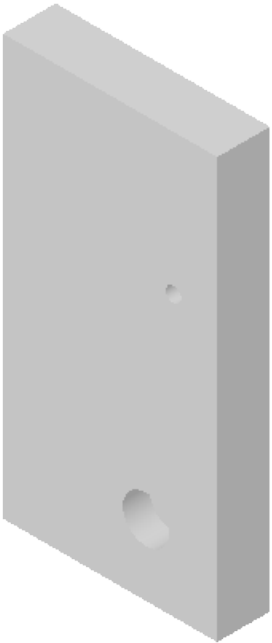
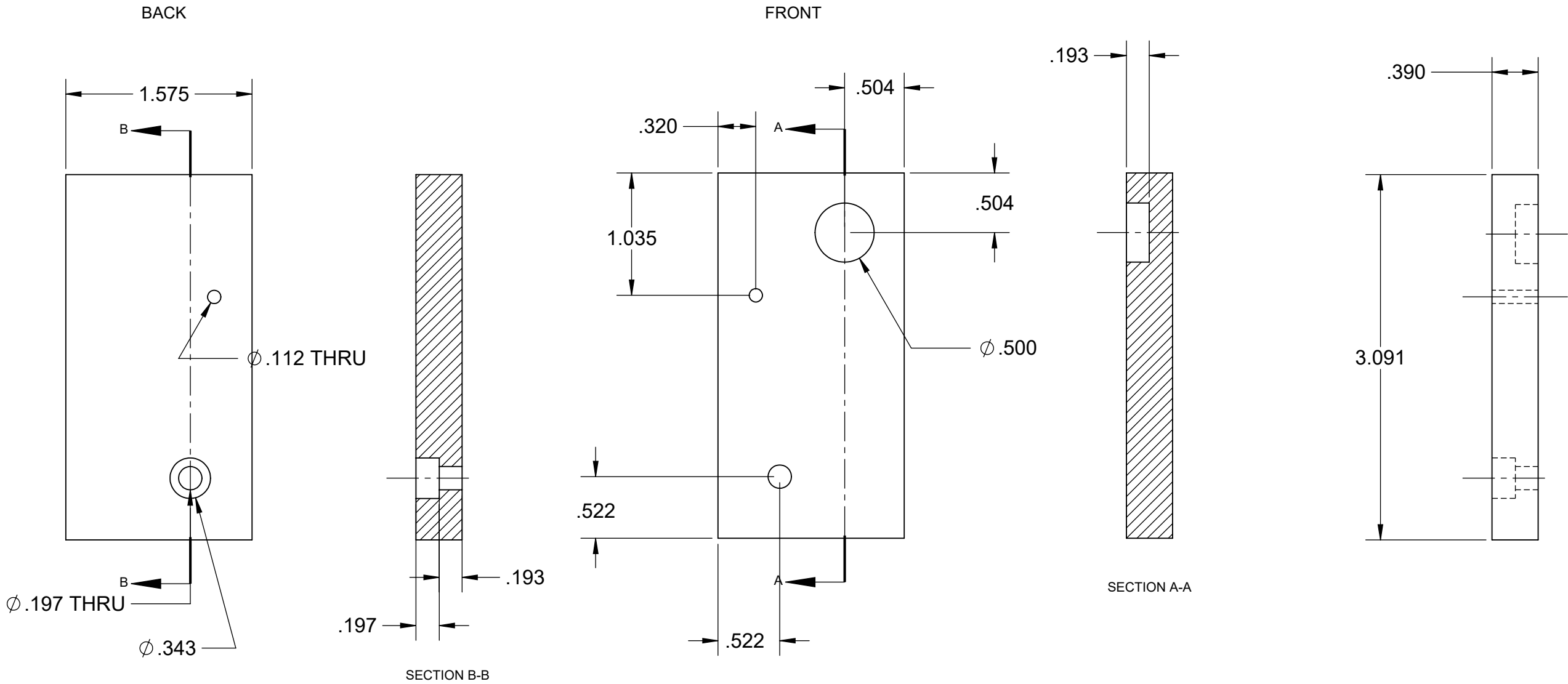


DETAIL C



Harvey Mudd College www.HMC.edu 301 Platt Blvd Claremont, CA 91711 ©2019 ALL RIGHTS RESERVED	TOLERANCES (EXCEPT AS NOTED)	NEXT ASSEMBLY	SHEET SIZE	PROJECT		
	DIMENSIONS IN INCHES[MM]	1	B	E4_Propulsion Lab Helper		
	DECIMAL: .xxx = ±.015	MATERIAL Carbon Fiber Nylon	DRAWING TITLE Tray			
	TYP EDGE BREAK: .005-.015	FINISH 3D Printed	DRAWN BY SQ	DATE 12/02/2025	SHEET 1 OF 1	
	ANGULAR: ±0.5°	PROJECT CODE Propulsion	DRAWING NUMBER T1			REV A

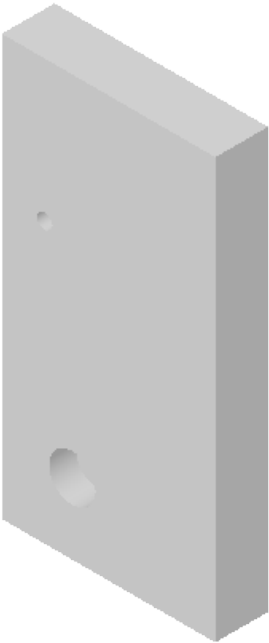
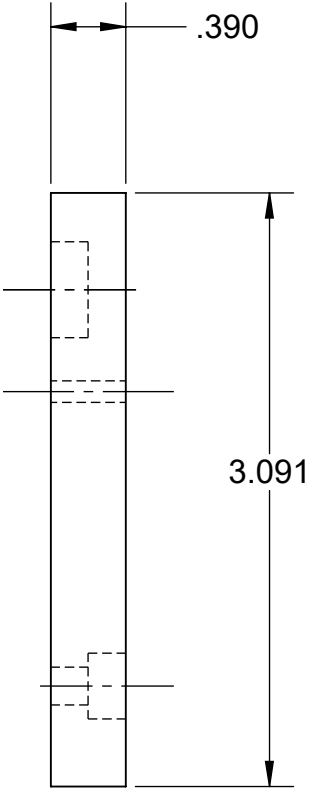
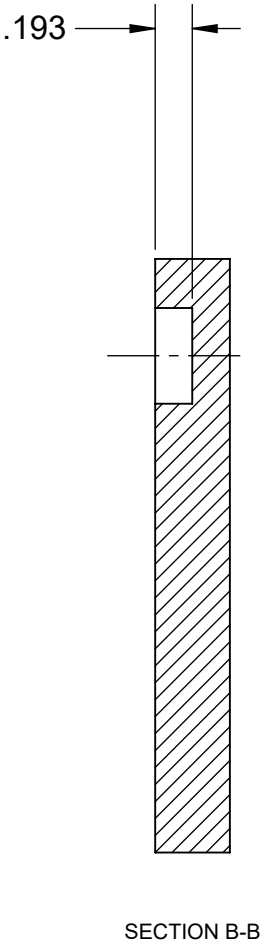
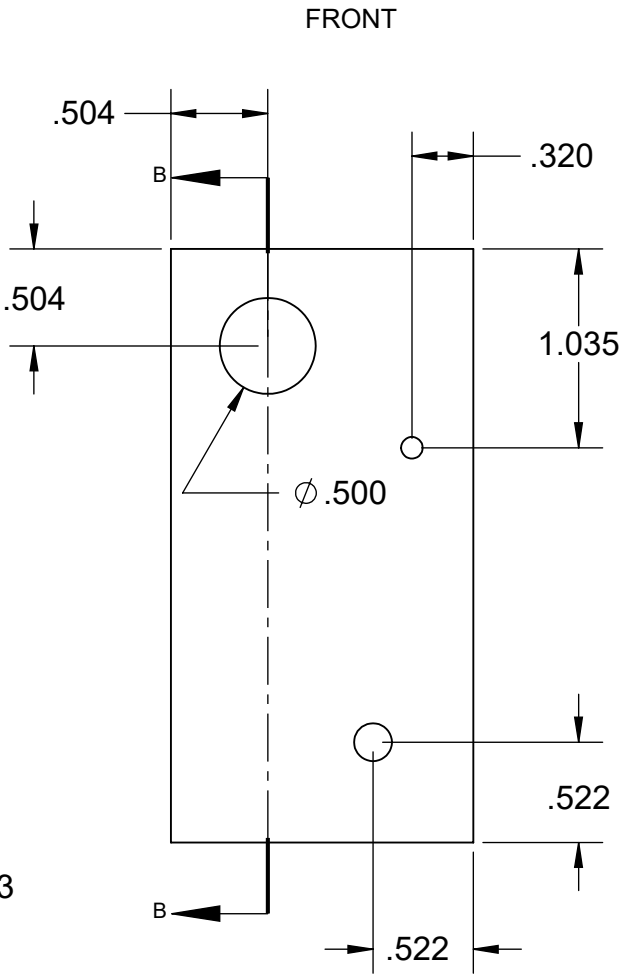
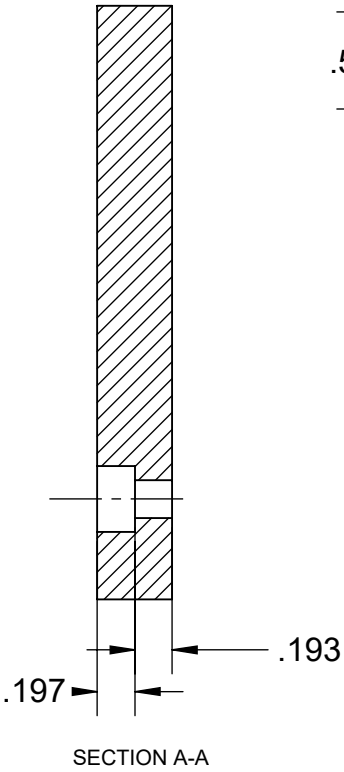
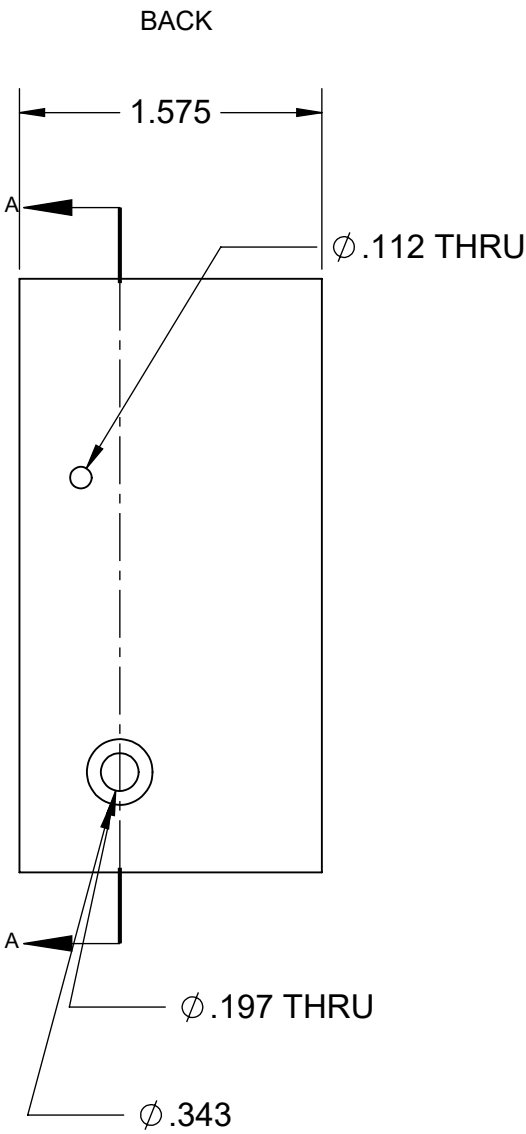
REV	DESCRIPTION	BY	DATE
A	INITIAL RELEASE	SQ	12/02/2025



Left Support; drawing created 11/30/2025 4:19:02 PM, last modified 12/2/2025 7:18:40 PM

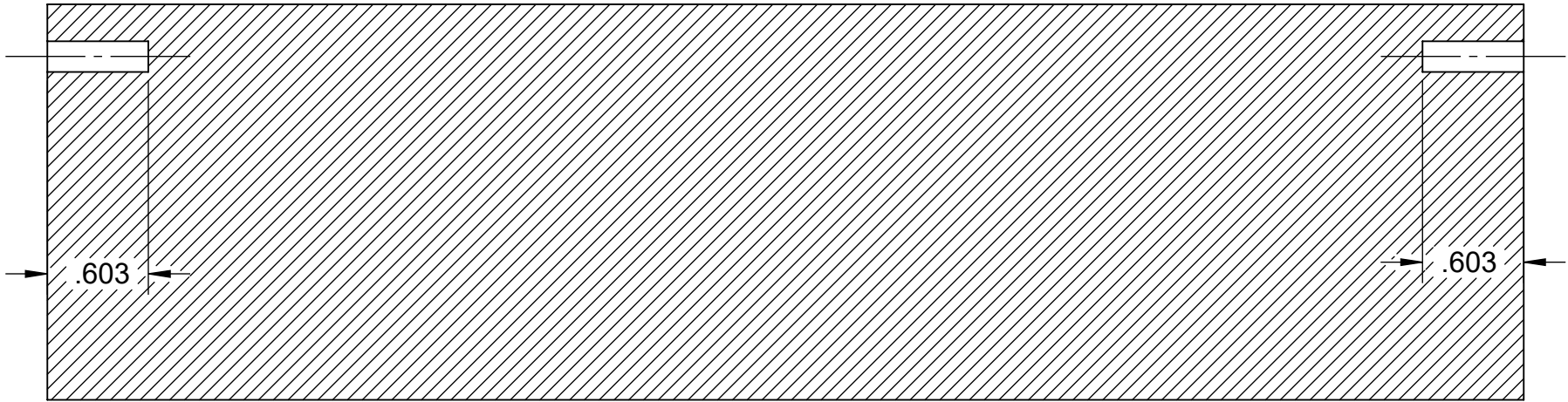
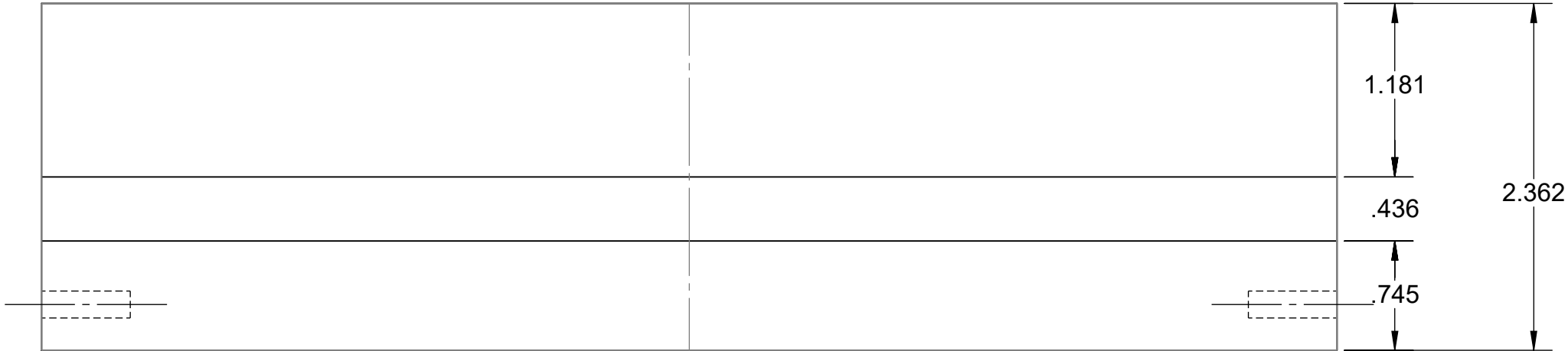
Harvey Mudd College www.HMC.edu 301 Platt Blvd Claremont, CA 91711 ©2019 ALL RIGHTS RESERVED	TOLERANCES (EXCEPT AS NOTED)	NEXT ASSEMBLY	SHEET SIZE	PROJECT		
	DIMENSIONS IN INCHES[IN]	1	B	E4_Propulsion Lab Helper		
	DECIMAL: .xxx = ±.005	MATERIAL Aluminum 3003	DRAWING TITLE Left Support			
	TYP EDGE BREAK: .005-.015	FINISH Polishing Disk	DRAWN BY SQ	DATE 12/02/2025	SHEET 1 OF 1	
	ANGULAR: ±0.5°	PROJECT CODE Propulsion	DRAWING NUMBER LS1			REV A

REV	DESCRIPTION	BY	DATE
A	INITIAL RELEASE	SQ	12/02/2025

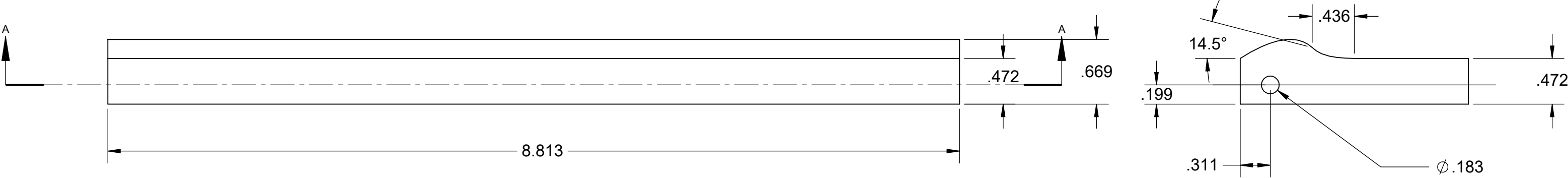
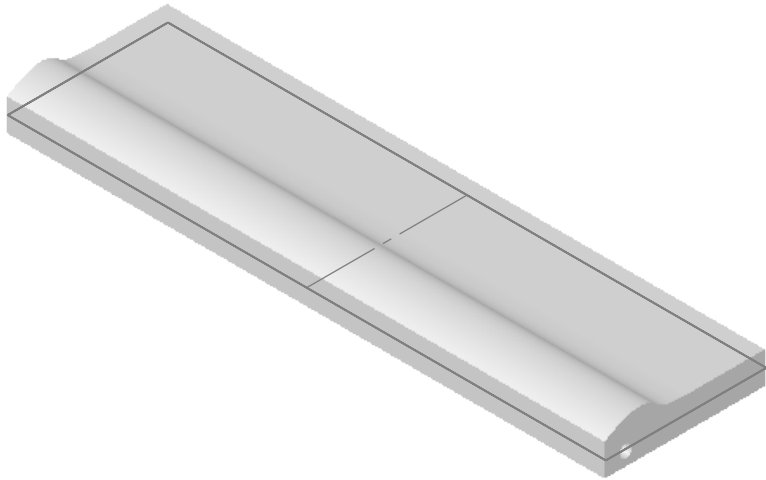


Harvey Mudd College www.HMC.edu 301 Platt Blvd Claremont, CA 91711 ©2019 ALL RIGHTS RESERVED	TOLERANCES (EXCEPT AS NOTED)	NEXT ASSEMBLY	SHEET SIZE	PROJECT		
	DIMENSIONS IN INCHES[IN]	1	B	E4_Propulsion Lab Helper		
	DECIMAL: .xxx = ±.005	MATERIAL	DRAWING TITLE			
	TYP EDGE BREAK: .005-.015	FINISH	DRAWN BY		DATE	SHEET
	ANGULAR: ±0.5°	PROJECT CODE	DRAWING NUMBER			REV

REV	DESCRIPTION	BY	DATE
A	INITIAL RELEASE	SQ	12/02/2025



SECTION A-A



Harvey
Mudd
College

www.HMC.edu
301 Platt Blvd
Claremont, CA 91711

©2019 ALL RIGHTS RESERVED

TOLERANCES
(EXCEPT AS NOTED)

DIMENSIONS IN
INCHES [IN]

DECIMAL:
.xxx = ±.015

TYP EDGE BREAK:

.005-.015

ANGULAR:
±0.5°

NEXT ASSEMBLY

1

MATERIAL

PLA Plastic

FINISH

3D Printed

PROJECT CODE

Propulsion

SHEET SIZE

B

PROJECT

E4_Propulsion Lab Helper

DRAWING TITLE

Press Board

DRAWN BY

SQ

DATE

12/02/2025

SHEET

1 OF 1

DRAWING NUMBER

PB1

REV

A

Bill of Materials					
	# in Solidworks OR # on part purchased				
Item	Part Number	Manufacturer	Quantity	Price per unit (\$)	Total cost for part (\$)
Aluminum 3003	-				0
Carbon Fiber - Nylon 3D filament	-				0
PLA 3D Filament	-		(gram estimate)		#VALUE!
Mini steel L-brackets	-		2		
Large steel L-brackets	-		2		0
3/16" x 1" Steel Dowel Pins	76388-H	Midwest Fastener	2	0.56	1.12
Ball Bearings	SSRS7R3DDF	New Departure Division & General Motors	2	2	4
4mm x 20mm Socket Cap Screws (0.70 pitch)	75615-N	Find-it-Fast	4	1.15	4.6
5mm x 12 mm Socket Cap Screws (0.80 pitch)	69646-F	Find-it-Fast	4	0.54	2.16
4-40 x 3/4" Zinc Slotted Round Head Machine Screws	62164-D	Midwest Fastener	2	0.11	0.22
4-40 Zinc Hex Nuts	62168-H	Midwest Fastener	2	0.1	0.2
8-32 Stop Nut	63782-B	Midwest Fastener	4	0.31	1.24
5mm Stainless Steel Metric Hex Nuts (0.80 pitch)	69623-C	Find-it-Fast	4	0.3	1.2
#6 1/8" Split Lock Washers	61941-A	Midwest Fastener	2	0.1	0.2
#10 3/16" Split Lock Washers	61943-C	Midwest Fastener	4	0.1	0.4
4mm Flat Washers	69582-B	Midwest Fastener	4	0.11	0.44
				Total cost for prototype:	#VALUE!

Base Process Router

Process	Machine	Comments
Cut aluminum stock.	Vertical bandsaw	Cut aluminum stock roughly to length, leaving about $\frac{1}{4}$ " excess to face on the mill.
Face aluminum stock to length.	Mill	Using the mill, face each side of the aluminum stock until it is the desired length. Use a square to ensure that all 4 corners are at 90 degrees.
Drill through holes for attaching lego base	Mill	Drill through holes using a 0.169" drill bit (or slightly larger if specific drill bit size is unavailable). Make sure to use a centering drill bit prior to drilling the hole to size.
Countersink for attaching lego base	Mill	Use a 0.28" diameter or slightly larger endmill to make countersinks to the specified depth on the drawing.
Drill through holes for attaching L brackets	Mill	Drill through holes using a 0.196" drill bit (or slightly larger if specific drill bit size is unavailable). Make sure to use a centering drill bit prior to drilling the hole to size.
Countersink for attaching L brackets	Mill	Use a 0.345" diameter or slightly larger endmill to make countersinks to the specified depth on the drawing.
Sanding corners and edges	Metal grinding wheel	Grind down all sharp edges and corners until they are safe to run a finger over them.
Debur and polish	File, handheld pneumatic grinder with polishing disk	Debur all holes and use the polishing disk to polish the flat surface of the aluminum.

Supports Process Router

Process	Machine	Comments
Cut aluminum stock.	Vertical bandsaw	Cut aluminum stock roughly to length, leaving about $\frac{1}{8}$ " excess to face on the mill.
Face aluminum stock to length.	Mill	Using the mill, face each side of the aluminum stock until it is the desired length. Use a square to ensure that all 4 corners are at 90 degrees.
Drill through holes for attaching L brackets	Mill	Drill through holes using a 0.196" drill bit (or slightly larger if specific drill bit size is unavailable). Make sure to use a centering drill bit prior to drilling the hole to size.
Countersink for attaching L brackets	Mill	Use a 0.345" diameter or slightly larger endmill to make countersinks to the specified depth on the drawing.
Mill hole for bearings	CNC Mill	Use a 0.5" drill bit and drill into supports by 0.175". Finish to specified length using a 0.5" diameter endmill. Alternatively, use a smaller diameter endmill and program the mill to cut in a clockwise swirl until the desired diameter is achieved.
Drill through holes for attaching press board support brackets	Mill	Drill through holes using a 0.110" diameter (or slightly larger if specific drill bit size is unavailable).
Sanding corners and edges	Metal grinding wheel	Grind down all sharp edges and corners until they are safe to run a finger over them.
Debur and polish	File, handheld pneumatic grinder with polishing disk	Debur all holes and use the polishing disk to polish the flat surface of the aluminum.